

Bootstrap Exercise — Responsive Webpage Using Grid System

Aim

To design a responsive web page using Bootstrap 5 that displays a title, three responsive columns, and styled buttons.

Theory

Bootstrap provides a 12-column grid system, utility classes, and responsive design principles. In this exercise, students will create a structured page layout that automatically adjusts for mobile, tablet, and desktop screens.

Key Bootstrap classes used:

- .container, .row, .col-md-* → Layout
- .text-center, .bg-*, .text-* → Text and background utilities
- .btn, .btn-* → Buttons
- .p-*, .m-* → Padding and margin utilities

Procedure

1. Create a new folder named 'bootstrap-ex1'.
2. Inside it, create a file named 'index.html'.
3. Add the Bootstrap CDN link in the <head>.
4. Design a simple page with three columns and styled buttons.
5. Test the page on different screen sizes (mobile, tablet, desktop).

Code: index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>Bootstrap Exercise</title>

  <!-- Bootstrap CSS -->
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
rel="stylesheet">
</head>

<body>
  <!-- Header Section -->
  <div class="container text-center mt-4">
    <h1 class="display-4 text-primary">Welcome to Bootstrap</h1>
    <p class="lead text-secondary">This is a responsive webpage built using the Bootstrap Grid
System.</p>
  </div>

  <!-- 3 Column Layout -->
  <div class="container mt-5">
    <div class="row text-center">
      <div class="col-md-4 col-sm-12 mb-4">
        <div class="p-4 bg-light border rounded">
          <h3 class="text-success">Column 1</h3>
          <p>Bootstrap makes responsive design simple and efficient.</p>
          <button class="btn btn-success">Read More</button>
        </div>
      </div>
      <div class="col-md-4 col-sm-12 mb-4">
        <div class="p-4 bg-light border rounded">
          <h3 class="text-info">Column 2</h3>
```

```

        <p>Use grid classes to adjust layouts on various screen sizes.</p>
        <button class="btn btn-info">Learn More</button>
    </div>
</div>
<div class="col-md-4 col-sm-12 mb-4">
    <div class="p-4 bg-light border rounded">
        <h3 class="text-danger">Column 3</h3>
        <p>Bootstrap helps you build beautiful designs faster!</p>
        <button class="btn btn-danger">Explore</button>
    </div>
</div>
</div>
</div>
</div>

<!-- Footer Section -->
<footer class="text-center mt-5 py-3 bg-dark text-white">
    &copy; 2025 Bootstrap Tutorial Exercise
</footer>

<!-- Bootstrap JS -->
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>

```

Expected Output

A responsive webpage with:

- A centered header
- Three boxes in one row (on desktop)
- Stacked columns on small screens
- Colorful buttons and consistent spacing

Learning Outcome

- Use the Bootstrap Grid System for layout design.
- Apply utility classes for spacing and text styling.
- Build responsive webpages that adapt to different screen sizes.
- Combine containers, rows, and columns effectively.

Reference

- Bootstrap Official Documentation: <https://getbootstrap.com/docs/5.3/layout/grid/>
- W3Schools Bootstrap 5 Tutorial: https://www.w3schools.com/bootstrap5/bootstrap_get_started.php